California Housing Price Prediction Report

This report presents a concise summary of exploratory data analysis (EDA), model building, evaluation metrics, and possible future improvements for a machine learning project focused on predicting California housing prices using Linear Regression.

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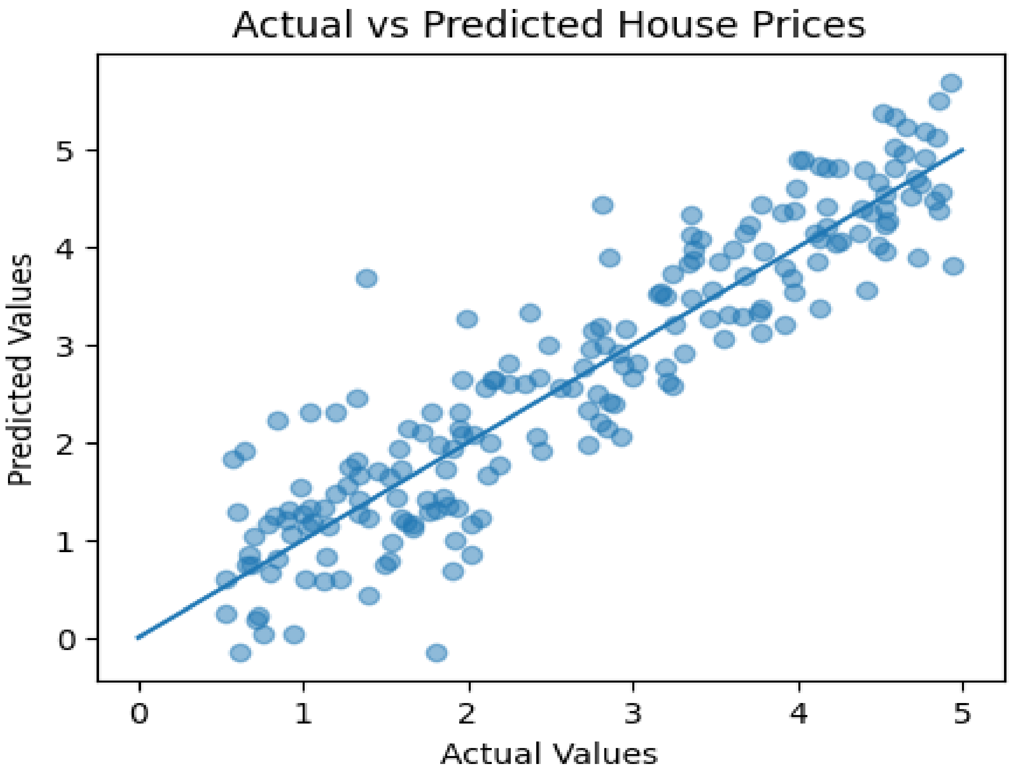
1. Exploratory Data Analysis (EDA)

The California Housing dataset contains housing-related attributes such as median income, average rooms, house age, population, and geographical coordinates. During the analysis, it was observed that median income had a strong relationship with house prices. Features related to population and occupancy showed wider distributions, which may introduce prediction variability.

Feature	Description
MedInc	Median Income
HouseAge	Average House Age
AveRooms	Average Rooms
Population	Population in Area
Latitude / Longitude	Location Information
MedHouseVal	Target Variable

2. Model Development

A Linear Regression model from the scikit-learn library was implemented. The dataset was divided into training and testing sets using an 80:20 split ratio. After training the model, predictions were generated on the testing data and evaluated using standard regression metrics.



3. Evaluation Metrics

Metric	Value
Mean Absolute Error (MAE)	0.5332
Mean Squared Error (MSE)	0.5559
R^2 Score	0.5758

The model achieved an R^2 score of 0.58, which indicates that the model explains a moderate portion of the variance in housing prices. Although Linear Regression provides a simple and interpretable baseline model, there is still room for performance improvement.

4. Improvement Ideas

• Perform feature scaling and normalization.
• Use feature engineering to capture non-linear relationships.
• Experiment with advanced algorithms such as Random Forest or XGBoost.
• Apply cross-validation for better model reliability.
• Detect and handle outliers to reduce prediction noise.

5. Conclusion

The project successfully demonstrated the use of machine learning techniques for house price prediction using Linear Regression. The workflow included data preparation, model training, evaluation, and visualization of results. Further improvements can enhance prediction accuracy and overall model robustness.